ADN-SDK 6.2.1.1 集成说明（iOS）

SDK 版本： 6.2.1.1
交付形态： `FLNAdSDK.xcframework`
最低系统： iOS 13（产品文档口径）
架构： 真机 `arm64`；模拟器 `arm64` / `x86_64`

本版本说明（相对 6.2.0.19）

- SDK 版本号：`6.2.1.1`（可通过 `[FLNAdSDKManager sdkVersion]` 确认）
- **对外广告 API 与 6.2.0.19 兼容**，无需修改接入代码
- Info.plist Scheme 白名单按下文推荐列表配置；已补充唤端 scheme，不加入 `com.` 形式
- 交付物：`FLNAdSDK.xcframework` + Demo（`FlowNetDemo`）；Demo 首页底部可查看当前 SDK 版本号

一、合规指引

接入前请阅读并按以下文档完成合规与隐私披露：

说明	链接
SDK 合规说明	https://adssp.innoads.cn/doc/protocol_doc/02sdk_developer_usage_compliance_specifications_20260306MK.html
App 合规说明	https://adssp.innoads.cn/doc/protocol_doc/01developer_compliance_guidelines20260309MK.html
App Store 隐私信息说明	https://adssp.innoads.cn/doc/pages/appst1/
易量跃动 ADN SDK 隐私政策	https://adssp.innoads.cn/doc/protocol_doc/03innoad_sdk_sprivacy_policy_20260309MK.html
易量跃动聚合广告 SDK 隐私政策	https://adssp.innoads.cn/doc/protocol_doc/05innoad_sdk_mediation_sprivacy_policy__20260309MK.html

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二、SDK 工程配置

1. 接入准备

- 创建应用和广告位，拿到 SDK 所需参数（`AppId`、各广告位 `posId`）
- SDK 支持的最低 iOS 版本：**iOS 13**
- 示例 Demo 随 SDK 一并提供

2. 本文用词说明

名词	含义
AppId	应用 Id，标识当前应用，需与广告位配套使用
posId	广告位 Id，标识广告位
FLNSplashAd	开屏广告
FLNNativeExpressAd	原生广告（模板渲染）
FLNUnifiedNativeAd	原生广告（自渲染）
FLNInterstitialAd	插屏广告
FLNRewardAd	激励视频广告

3. 支持的广告类型

开屏、激励视频、插屏、信息流模板、信息流自渲染。

4. 接入 SDK

1. 将 `FLNAdSDK.xcframework` 拖入 Xcode 工程
2. **General → Frameworks, Libraries, and Embedded Content** 中设为 **Embed & Sign**
3. 建议链接以下系统库：
    - `AdSupport.framework`
    - `AVFoundation.framework`
    - `CoreAudio.framework`
    - `CoreServices.framework`
    - `AppTrackingTransparency.framework`（iOS 14+ ATT）

5. Info.plist：推荐设置 Scheme 白名单

配置白名单可提高广告收益：

```
<key>LSApplicationQueriesSchemes</key>
<array>
  <string>openapp.jdmobile</string>
  <string>openjd</string>
  <string>jdmobile</string>
  <string>vipshop</string>
  <string>dianping</string>
  <string>imeituan</string>
  <string>taobao</string>
  <string>tmall</string>
  <string>wireless1688</string>
  <string>tbopen</string>
  <string>taobaolite</string>
  <string>taobaoliveshare</string>
  <string>alipays</string>
  <string>pddopen</string>
  <string>pinduoduo</string>
  <string>weixin</string>
  <string>weixinULAPI</string>
  <string>weixinURLParamsAPI</string>
  <string>snssdk141</string>
  <string>ctrip</string>
  <string>suning</string>
  <string>qunariphone</string>
  <string>QunarAlipay</string>
  <string>qunaraphone</string>
  <string>kaola</string>
  <string>agoda</string>
  <string>jhs</string>
  <string>yhd</string>
  <string>wbmain</string>
  <string>xhsdiscover</string>
  <string>douyin</string>
  <string>mogujie</string>
  <string>koubei</string>
  <string>eleme</string>
  <string>youku</string>
  <string>didicommon</string>
  <string>OneTravel</string>
  <string>snssdk1112</string>
  <string>snssdk1128</string>
  <string>miguvideo</string>
  <string>kfcapplinkurl</string>
  <string>iqiyi</string>
  <string>uclink</string>
  <string>baiduboxapp</string>
  <string>iosamap</string>
  <string>tongyi</string>
</array>
```

6. iOS 14 适配

由于 iOS 14 对权限和隐私有调整，且与广告业务关系较大，请按以下步骤适配。未适配一般不会导致崩溃，但会在一定程度上影响广告收入。

1. 应用编译环境建议升级至 **Xcode 14.0** 及以上
2. 设置 IDFA 权限（App Tracking Transparency）

6.1 获取 App Tracking Transparency 授权

自 **iOS 14.5** 起，在应用调用 ATT 向用户请求跟踪授权之前，IDFA 将不可用。

① 在 Info.plist 添加 `NSUserTrackingUsageDescription`

```
<key>NSUserTrackingUsageDescription</key>
<string>获取标记权限向您提供更优质、安全的个性化服务及内容，未经同意我们不会用于其他目的；开启后，您也可以前往系统“设置-隐私 ”中随时关闭</string>
```

文案建议（二选一）：

- 获取标记权限向您提供更优质、安全的个性化服务及内容，未经同意我们不会用于其他目的；开启后，您也可以前往系统“设置-隐私 ”中随时关闭。
- 获取IDFA标记权限向您提供更优质、安全的个性化服务及内容；开启后，您也可以前往系统“设置-隐私 ”中随时关闭。

② 向用户申请权限

```
#import <AppTrackingTransparency/AppTrackingTransparency.h>
#import <AdSupport/AdSupport.h>

- (void)requestIDFA {
    if (@available(iOS 14, *)) {
        [ATTrackingManager requestTrackingAuthorizationWithCompletionHandler:^(ATTrackingManagerAuthorizationStatus status) {
            // 无需对授权状态做业务处理
        }];
    }
}
```

③ iOS 15 不弹出授权窗的解决方法

根据官方说明，可将权限申请放在 `applicationDidBecomeActive:`（建议在用户同意隐私协议后再调用）：

```
- (void)applicationDidBecomeActive:(UIApplication *)application {
    // 用户同意协议后获取
    [self requestIDFA];
}
```

参考：https://developer.apple.com/forums/thread/690607

三、初始化与广告接入

通用约定

项	说明
初始化	必须先调用 `startWithConfig:completionHandler:`，再请求广告
广告位	各广告类使用 `initWithPosId:`，`posId` 由后台下发
超时	开屏 / 激励 / 模板支持 `timeout`，**必须在 `loadAd` 前设置**
就绪	开屏 / 激励 / 插屏 / 模板：`isReady == YES` 后再展示；自渲染请判断单条 `dataObject.isAdValid`（已移除 `FLNUnifiedNativeAd.isReady`）
eCPM	类型为 **`float`**，单位：**分**（可为小数）；无权限或异常多为 `-1`
错误码	对外 `NSError.domain` 一般为 `kFLNErrorDomain`，详见 `FLNErrorCodeHeader.h`

竞价通知（FLNAdProtocol）

开屏、激励、插屏、模板、自渲染数据对象均实现 `FLNAdProtocol`。各 value 请传 **`NSString`**，价格单位为分（可为小数）。

winner_adn（胜出方 ADN / 输给哪家）：

取值	含义
`unknown`	未知
`csj`	穿山甲
`gdt`	优量汇
`bd`	百度联盟/百青藤
`ks`	快手联盟
`sigmob`	Sigmob
`mtg`	Mintegral
`inmobi`	InMobi
`other`	其它

loss_reason（竞败原因，仅竞败）：

取值	含义
`100`	返回超时
`201`	缺少竞价价格
`202`	出价低于底价
`203`	竞价失败
`900`	其他原因

竞胜之后调用，需在广告 show 之前调用：

```
[ad sendWinNotificationWithInfo:@{
    @"second_price": @"80",   // 次高价 / 结算价，NSString
    @"winner_adn": @"other"   // 见上方 winner_adn 说明
}];
```

竞败之后或未参竞调用：

```
[ad sendLossNotificationWithInfo:@{
    @"winner_price": @"120",  // 竞赢方最高价，NSString
    @"winner_adn": @"csj",    // 见上方 winner_adn 说明
    @"loss_reason": @"203"    // 见上方 loss_reason 说明
}];
```

> 自渲染场景对 `FLNUnifiedNativeAdDataObject` 调用即可。

对外错误码（FLNErrorCode）

```
#import <FLNAdSDK/FLNErrorCodeHeader.h>
// error.domain 一般为 kFLNErrorDomain
// code 在 1000~5999 为 SDK 自有码，其它值可能为服务端业务码
```

分段	码	含义
1xxx 客户端/状态	`1001`	入参非法（缺 posId/appId、非法 window/VC 等）
	`1002`	重复发起加载
	`1003`	广告已失效（缓存过期 / 已曝光消费）
	`1004`	广告未 ready
2xxx 初始化	`2001`	SDK 初始化失败
3xxx 加载	`3001`	无填充
	`3002`	加载超时
	`3003`	网络错误
	`3004`	加载失败兜底
4xxx 素材	`4001`	视频素材加载/准备失败
	`4002`	图片素材加载失败
	`4003`	模板/创意渲染失败
5xxx 展示	`5001`	视频播放失败
	`5002`	落地页 URL 无效
	`5003`	曝光未达阈值
	`5004`	展示失败兜底

5.1 初始化

```
#import <FLNAdSDK/FLNAdSDKManager.h>
#import <FLNAdSDK/FLNInitConfig.h>

- (BOOL)application:(UIApplication *)application didFinishLaunchingWithOptions:(NSDictionary *)launchOptions {
    self.window = [[UIWindow alloc] initWithFrame:[[UIScreen mainScreen] bounds]];
    UIViewController *rootVC = [[ViewController alloc] init];
    UINavigationController *navigationVC = [[UINavigationController alloc] initWithRootViewController:rootVC];
    self.window.rootViewController = navigationVC;
    navigationVC.navigationBar.hidden = YES;
    [self.window makeKeyAndVisible];
    [self initAdSDK];
    return YES;
}

- (void)initAdSDK {
    FLNInitConfig *config = [FLNInitConfig new];
    config.appId = @"your_app_id";
    config.userId = @"your_user_id"; // 可选
    [[FLNAdSDKManager sharedInstance] startWithConfig:config
                                    completionHandler:^(BOOL success, NSError * _Nullable error) {
        if (!success) {
            NSLog(@"FLN init failed: %@", error);
            return;
        }
        NSLog(@"FLN init success, sdkVersion=%@", [FLNAdSDKManager sdkVersion]);
    }];
}
```

5.2 开屏广告

```
#import <FLNAdSDK/FLNSplashAd.h>

@interface SplashViewController () <FLNSplashAdDelegate>
@property (nonatomic, strong) FLNSplashAd *splashAd;
@end

@implementation SplashViewController

- (void)viewDidLoad {
    [super viewDidLoad];
    self.splashAd = [[FLNSplashAd alloc] initWithPosId:@"your_splash_pos_id"];
    self.splashAd.delegate = self;
    self.splashAd.timeout = 5; // 必须在 load 前设置
}

- (void)loadAd {
    [self.splashAd loadAd];
}

- (void)showAd {
    if (![self.splashAd isReady]) {
        return;
    }
    UIWindow *window = self.view.window; // 或取 keyWindow / 前台 Scene 的 window

    // 全屏展示
    [self.splashAd showAdInWindow:window];

    // 或：底部贴宿主自定义视图（如 App Logo 区）
    // UIView *bottomView = ...; // 高度取该 view 自身高度，且不超过屏高 25%；传 nil 等同全屏
    // [self.splashAd showAdInWindow:window withBottomView:bottomView];
}

#pragma mark - FLNSplashAdDelegate

- (void)splashAdDidLoad:(FLNSplashAd *)splashAd {
    NSLog(@"splash load success, eCPM=%.2f", splashAd.eCPM);
    // 竞价场景：竞胜后、show 前调用 sendWinNotificationWithInfo:
}

- (void)splashAdLoadFail:(FLNSplashAd *)splashAd error:(NSError *)error {
    NSLog(@"splash load fail: %@", error);
}

- (void)splashAdDidShow:(FLNSplashAd *)splashAd {}
- (void)splashAdShowFailed:(FLNSplashAd *)splashAd withError:(NSError *)error {}
- (void)splashAdClicked:(FLNSplashAd *)splashAd {}
- (void)splashAdClosed:(FLNSplashAd *)splashAd {}
- (void)splashAdSkip:(FLNSplashAd *)splashAd time:(NSInteger)time {}

@end
```

5.3 信息流模板广告

```
#import <FLNAdSDK/FLNNativeExpressAd.h>
#import <FLNAdSDK/FLNNativeExpressAdView.h>

@interface NativeExpressAdViewController () <FLNNativeExpressAdDelegate, UITableViewDelegate, UITableViewDataSource>
@property (nonatomic, strong) FLNNativeExpressAd *nativeExpressAd;
@property (nonatomic, strong) NSMutableArray<FLNNativeExpressAdView *> *adPool;
@property (nonatomic, strong) UITableView *tableView;
@property (nonatomic, strong) NSMutableArray *data;
@end

@implementation NativeExpressAdViewController

- (void)viewDidLoad {
    [super viewDidLoad];
    // ... 准备 tableView 与内容占位（如 "__ad__"） ...

    CGFloat width = CGRectGetWidth(self.view.bounds);
    if (width < 1) {
        width = CGRectGetWidth(UIScreen.mainScreen.bounds);
    }
    // 使用 width 初始化（不再使用 adSize）
    self.nativeExpressAd = [[FLNNativeExpressAd alloc] initWithPosId:@"your_express_pos_id" width:width];
    self.nativeExpressAd.delegate = self;
    self.nativeExpressAd.timeout = 5; // 可选，须在 load 前设置
    [self.nativeExpressAd loadAd:1];  // 一次请求条数
}

#pragma mark - FLNNativeExpressAdDelegate

- (void)nativeExpressAdDidLoad:(FLNNativeExpressAd *)nativeExpressAd {
    NSLog(@"广告 Load");
}

- (void)nativeExpressAdFailToLoad:(FLNNativeExpressAd *)nativeExpressAd error:(NSError *)error
{
    NSLog(@"广告加载失败: %@", error);
}

- (void)nativeExpressAdViewRenderSuccess:(FLNNativeExpressAd *)nativeExpressAd
                     nativeExpressAdView:(NSArray<__kindof FLNNativeExpressAdView *> *)nativeExpressAdViews {
    for (FLNNativeExpressAdView *adView in nativeExpressAdViews) {
        [self.adPool addObject:adView];
        // 可取 adView.eCPM（float，单位：分）参与竞价
    }
    [self.tableView reloadData];
}

- (void)nativeExpressAdViewRenderFail:(FLNNativeExpressAd *)nativeExpressAd error:(NSError *)error {
    NSLog(@"广告 RenderFail");
}

- (void)nativeExpressAdViewDidShow:(FLNNativeExpressAd *)nativeExpressAd
               nativeExpressAdView:(FLNNativeExpressAdView *)nativeExpressAdView {}

- (void)nativeExpressAdViewClicked:(FLNNativeExpressAd *)nativeExpressAd
               nativeExpressAdView:(FLNNativeExpressAdView *)nativeExpressAdView {}

- (void)nativeExpressAdViewClosed:(FLNNativeExpressAd *)nativeExpressAd
              nativeExpressAdView:(FLNNativeExpressAdView *)nativeExpressAdView {
    [self.adPool removeObject:nativeExpressAdView];
    [self.tableView reloadData];
}

@end
```

列表中将渲染成功的 `FLNNativeExpressAdView` 加入 Cell / 容器即可。关闭后从父视图移除并刷新列表。

5.4 信息流自渲染

```
#import <FLNAdSDK/FLNUnifiedNativeAd.h>
#import <FLNAdSDK/FLNUnifiedNativeAdView.h>
#import <FLNAdSDK/FLNUnifiedNativeAdDataObject.h>

@interface UnifiedNativeAdViewController () <FLNUnifiedNativeAdDelegate, FLNUnifiedNativeAdViewDelegate>
@property (nonatomic, strong) FLNUnifiedNativeAd *nativeAd;
@property (nonatomic, strong) NSMutableArray<FLNUnifiedNativeAdView *> *adViews;
@end

@implementation UnifiedNativeAdViewController

- (void)viewDidLoad {
    [super viewDidLoad];
    self.adViews = [NSMutableArray array];
    self.nativeAd = [[FLNUnifiedNativeAd alloc] initWithPosId:@"your_unified_pos_id"];
    self.nativeAd.delegate = self;
}

- (void)loadAd {
    [self.nativeAd loadAd];
}

- (void)clearAllAds {
    for (FLNUnifiedNativeAdView *v in [self.adViews copy]) {
        [v unregisterDataObject];
        v.delegate = nil;
        [v removeFromSuperview];
    }
    [self.adViews removeAllObjects];
}

- (void)buildAdViewWithData:(FLNUnifiedNativeAdDataObject *)data {
    if (!data.isAdValid) {
        return; // 过期 / 已展示等无效数据勿展示，否则影响计费
    }

    FLNUnifiedNativeAdView *adView = [[FLNUnifiedNativeAdView alloc] init];
    adView.viewController = self; // 用于弹出落地页，一般为当前 VC
    adView.delegate = self;
    [self.adViews addObject:adView];

    // 自建 UI：title / desc / imageUrl / iconUrl / buttonText / callToAction / videoUrl / videoImageUrl 等
    UIImageView *adImageView = [[UIImageView alloc] init];
    UIImageView *iconImageView = [[UIImageView alloc] init];
    UILabel *titleLabel = [[UILabel alloc] init];
    titleLabel.text = data.title.length ? data.title : @"广告";
    UILabel *descLabel = [[UILabel alloc] init];
    descLabel.text = data.desc ?: @"";
    UIButton *ctaButton = [UIButton buttonWithType:UIButtonTypeSystem];
    NSString *cta = data.buttonText.length ? data.buttonText
                  : (data.callToAction.length ? data.callToAction : @"查看详情");
    [ctaButton setTitle:cta forState:UIControlStateNormal];

    // 视频封面：data.videoImageUrl；时长：data.videoDuration（秒）
    // ... 布局并加载图片 ...

    NSArray<UIView *> *clickableViews = @[adImageView, iconImageView, titleLabel, descLabel, ctaButton];
    // register 与 unregister 须成对；避免重复 register；isAdValid==NO 时 register 不生效
    [adView registerDataObject:data clickableViews:clickableViews];
}

#pragma mark - FLNUnifiedNativeAdDelegate

- (void)fln_unifiedNativeAdLoaded:(NSArray<FLNUnifiedNativeAdDataObject *> *)unifiedNativeAdDataObjects
                            error:(NSError *)error {
    if (error) {
        return;
    }
    [self clearAllAds];
    for (FLNUnifiedNativeAdDataObject *obj in unifiedNativeAdDataObjects) {
        [self buildAdViewWithData:obj];
        // 竞价：对 dataObject 调用 sendWin / sendLoss
    }
}

#pragma mark - FLNUnifiedNativeAdViewDelegate

- (void)fln_unifiedNativeAdViewDidExpose:(FLNUnifiedNativeAdView *)unifiedNativeAdView {}
- (void)fln_unifiedNativeAdViewDidClick:(FLNUnifiedNativeAdView *)unifiedNativeAdView {}
- (void)fln_unifiedNativeAdDetailViewClosed:(FLNUnifiedNativeAdView *)unifiedNativeAdView {}
- (void)fln_unifiedNativeAdDetailViewWillPresentScreen:(FLNUnifiedNativeAdView *)unifiedNativeAdView {}

- (void)dealloc {
    self.nativeAd.delegate = nil;
    [self clearAllAds];
}

@end
```

自渲染注意：

- 展示前判断 `dataObject.isAdValid`（缓存过期 **或** 已展示过 均返回 NO）
- `registerDataObject:clickableViews:` 与 `unregisterDataObject` 成对调用；无效数据 register 直接不生效
- TableView / CollectionView 复用时，在 `prepareForReuse` 等时机 `unregisterDataObject`
- 常用字段：`title`、`desc`、`imageUrl`、`iconUrl`、`videoUrl`、`videoImageUrl`（视频封面）、`videoDuration`（秒）、`eCPM`（float）、`buttonText`、`callToAction`、`isAppAd`、`appPrice`

5.5 插屏广告

```
#import <FLNAdSDK/FLNInterstitialAd.h>

@interface InterstitialAdViewController () <FLNInterstitialAdDelegate>
@property (nonatomic, strong) FLNInterstitialAd *interstitialAd;
@end

@implementation InterstitialAdViewController

- (void)viewDidLoad {
    [super viewDidLoad];
    self.interstitialAd = [[FLNInterstitialAd alloc] initWithPosId:@"your_interstitial_pos_id"];
    self.interstitialAd.delegate = self;
}

- (void)loadAd {
    [self.interstitialAd loadAd];
}

- (void)showAd {
    if (![self.interstitialAd isReady]) {
        return;
    }
    UIViewController *root = self.navigationController ?: self;
    [self.interstitialAd presentAdFromRootViewController:root];
}

#pragma mark - FLNInterstitialAdDelegate

- (void)interstitialAdDidLoad:(FLNInterstitialAd *)interstitialAd {
    NSLog(@"eCPM=%.2f", interstitialAd.eCPM);
}

- (void)interstitialAd:(FLNInterstitialAd *)interstitialAd didFailWithError:(NSError *)error {}
- (void)interstitialAdDidShow:(FLNInterstitialAd *)interstitial {}
- (void)interstitialAdShowFailed:(FLNInterstitialAd *)interstitialAd error:(NSError *)error {}
- (void)interstitialAdDidClose:(FLNInterstitialAd *)interstitialAd {}
- (void)interstitialAdDidClicked:(FLNInterstitialAd *)interstitialAd {}

@end
```

5.6 激励视频广告

```
#import <FLNAdSDK/FLNRewardAd.h>

@interface RewardAdViewController () <FLNRewardAdDelegate>
@property (nonatomic, strong) FLNRewardAd *rewardAd;
@end

@implementation RewardAdViewController

- (void)viewDidLoad {
    [super viewDidLoad];
    self.rewardAd = [[FLNRewardAd alloc] initWithPosId:@"your_reward_pos_id"];
    self.rewardAd.delegate = self;
    self.rewardAd.timeout = 5;
    self.rewardAd.videoMuted = NO; // YES 为静音播放
}

- (void)loadAd {
    [self.rewardAd loadAd];
}

- (void)showAd {
    if (![self.rewardAd isReady]) {
        return;
    }
    UIViewController *root = self.navigationController ?: self;
    [self.rewardAd presentAdFromRootViewController:root];
}

#pragma mark - FLNRewardAdDelegate

- (void)rewardVideoAdDidLoad:(FLNRewardAd *)rewardedAd {
    NSLog(@"eCPM=%.2f", rewardedAd.eCPM);
}

- (void)rewardVideoAd:(FLNRewardAd *)rewardedAd didFailWithError:(NSError *)error {}
- (void)rewardVideoAdReadyToPlay:(FLNRewardAd *)rewardedAd {}
- (void)rewardVideoAdDidPlayStart:(FLNRewardAd *)rewardedAd {}
- (void)rewardVideoAdDidPlayFinish:(FLNRewardAd *)rewardedAd {}
- (void)rewardVideoAdDidShow:(FLNRewardAd *)rewardedAd {}
- (void)rewardVideoAdShowFailed:(FLNRewardAd *)rewardedAd error:(NSError *)error {}
- (void)rewardVideoAdDidClose:(FLNRewardAd *)rewardedAd {}
- (void)rewardVideoAdDidClicked:(FLNRewardAd *)rewardedAd {}

/// 奖励有效回调：在此给用户发放奖励
- (void)rewardVideoAdDidRewardEffective:(FLNRewardAd *)rewardedAd info:(NSDictionary *)info {
    NSLog(@"reward info: %@", info);
}

@end
```

发奖请以 `rewardVideoAdDidRewardEffective:info:` 为准。

四、注意事项

1. **先初始化再请求广告**
2. 开屏使用真实可用的 `UIWindow`；激励 / 插屏使用当前可 present 的 ViewController
3. 自渲染未 register、广告过期 / 已展示或无效数据会导致不计费 / 点击无效
4. 可通过 `[FLNAdSDKManager sdkVersion]` 确认版本（本包为 `6.2.1.1`）
